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High-salt d(CpGpCpG), a left-handed Z' DNA double helix.

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Abstract

The DNA tetramer d(CpGpCpG) or CGCG crystallizes from high-salt solution as a left-handed double helix, the Z' helix. Its structure differs from that of the other known left-handed helix, Z-DNA, by a C1'-exo sugar pucker at deoxyguanosines rather than C3'-endo, and they represent two alternative solutions to the same steric constraint arising from the syn glycosyl bond orientation. The apparent molecular weight for the Z to Z' transition in going from intermediate to high salt is substitution of a bound anion for water at guanine amino groups, consequent charge repulsion of anions and backbone phosphates.

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